



IILM University

Python Full Stack Development Internship

Capstone Project: Markdown to HTML Converter

Presented by: Krish Gupta

Roll No: 2410031425

Program : B.Tech-CSE

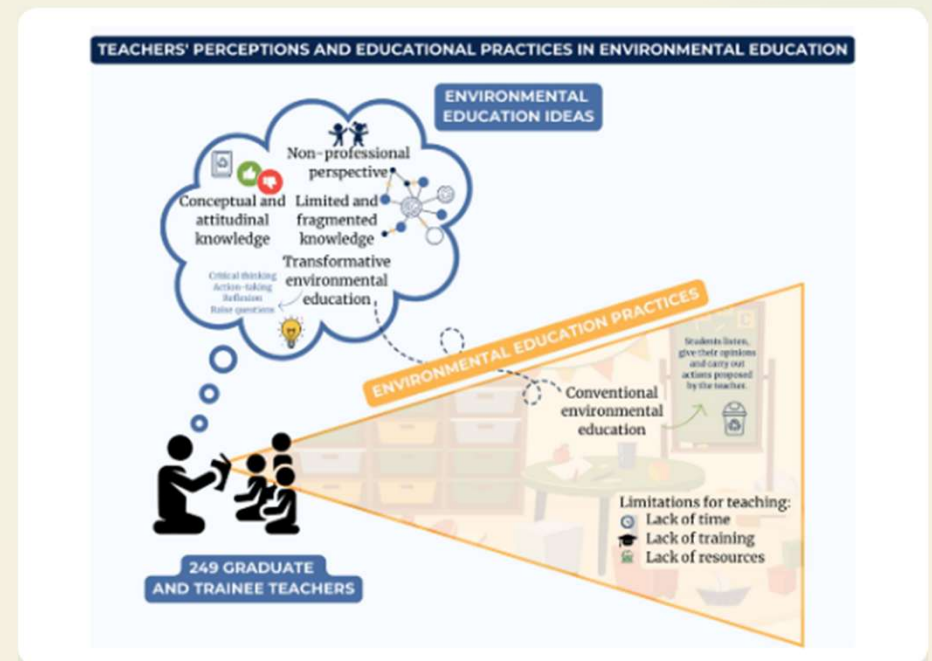
Semester: 5th

IILM University, Greater Noida

Organization: Eduskills Academy

Organization & Internship Overview

- **Organization:** EduSkills Academy & AICTE.
- **Domain:** EdTech & Skill Development focusing on industry-relevant skills.
- **Duration:** 8-Week Virtual Internship (June 2026 - August 2026).
- **Objective:** Bridge the gap between academic knowledge and real-world application through hands-on full stack development.



8-Week Learning Journey

Weeks 3-4

React fundamentals, Hooks, state management, and API integration.

Weeks 7-8

Testing, deployment strategies, and Capstone Project development.

Weeks 1-2

Python Fundamentals, OOP, and FastAPI backend setup with Pydantic validation.

Weeks 5-6

SQLAlchemy DB integration, advanced React patterns, and routing.



Capstone Project

Markdown to HTML Converter

Problem Statement & Objectives

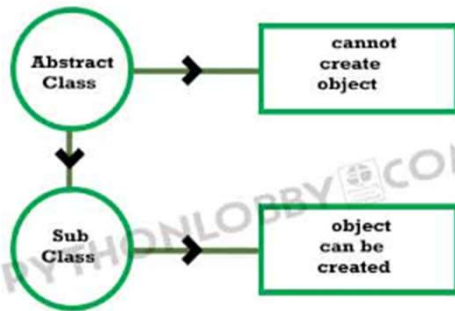
The Problem

There is a constant need for utilities to automate the translation of plain text (Markdown) into equivalent HTML code for web display. This requires a robust parser that handles headers, emphasis, inline code, and hyperlinks seamlessly.

Project Scope

Develop a Python-based sequential processing pipeline utilizing regular expressions to efficiently parse standard Markdown syntax and output valid, semantically correct HTML without unintended tag overlaps.

Core Technologies Utilized



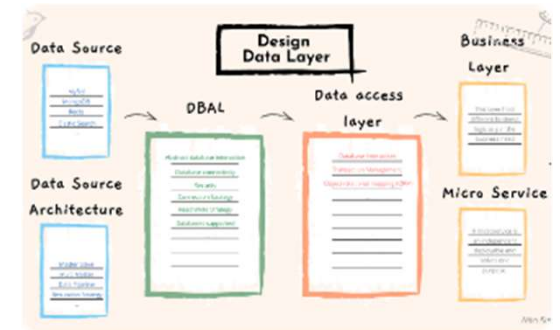
Python Backend

String manipulation, regex (`re` module), and logic formulation.



React & FastAPI

Full stack knowledge applied for robust API design and UI integration.

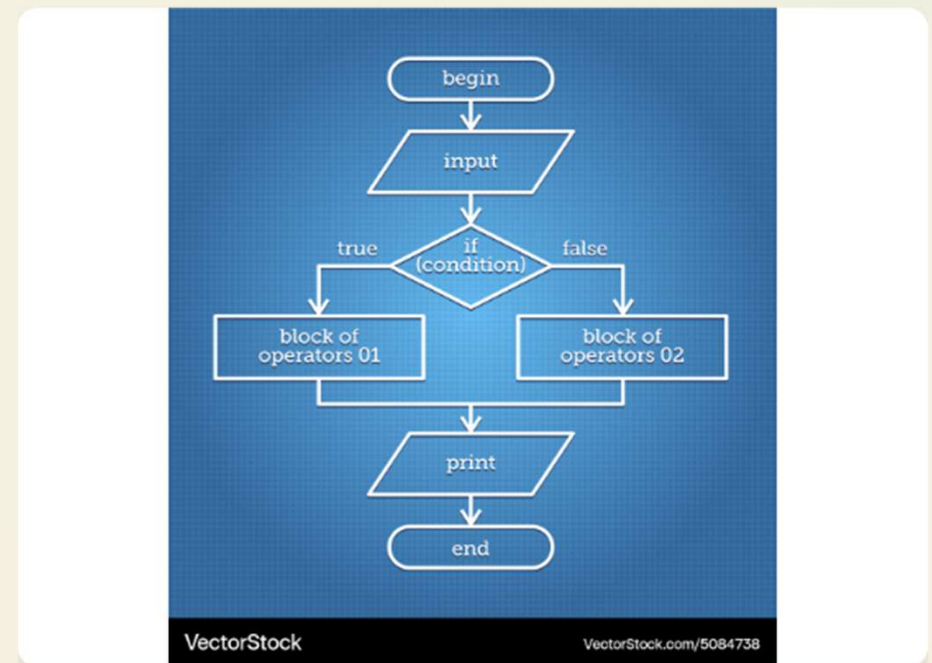


SQLAlchemy

Relational database integration and ORM usage during the learning modules.

Methodology & Pipeline

- **Header Parsing:** Identification of hash `#` prefixes to generate `h1` to `h6` HTML tags.
- **Emphasis Logic:** Regex non-greedy matching to convert asterisks to `strong` and `em` elements.
- **Inline Code:** Capturing text between backticks to wrap in `code` tags.
- **Hyperlinks:** Using capture groups ``\[((^\\)]+\\)\[((^)]+\\)`` to extract text and URLs.
- **Sequential Flow:** Processing in a specific order (Links > Bold > Italic > Code) to prevent formatting conflicts.



Challenges & Solutions



Overlapping Regex Matches Accurately handling nested Markdown syntax without unintended substitutions.

Solution: Crafted specific regex patterns using non-greedy matching.



Order of Operations Parsing italic text before bold text caused bold markers to break. *Solution: Enforced a strict sequential pipeline processing Links, then Bold, then Italics.*



Semantic Validation Ensuring the generated HTML is valid across different browser renderers. *Solution: Rigorous testing against expected HTML structures using unit tests.*

Key Internship Outcomes

8

Weeks of Training

Hands-On Experience

Successfully transitioned from core Python fundamentals to advanced state management in React and database integration. The Capstone Project solidified problem-solving skills, regex proficiency, and the ability to build functional, sequential data pipelines from scratch.

Thank You

Krish Gupta | Roll No:2410031425 | B.Tech CSE | IILM University